

Part I

Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 Fill in the following blanks with the correct choice of the words from this list:

Increasing, decreasing, positive, negative, concave up, concave down

- (a) If you know that $f''(x) > 0$, for all x in some open interval I , then you know $f'(x)$ is _____ and $f(x)$ is _____, for all x in I .
- (b) If you know $g'(x) < 0$ and decreasing, for all x in some open interval I , then you know $g(x)$ is _____ and _____, for all x in I .
- (c) If you know $h(x)$ is positive, increasing, and concave down, for all x in some open interval I , then you know $h'(x)$ is _____ and _____ and that $h''(x)$ is _____, for all x in I .

Problem 2

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Problem 2

Problem 2 Suppose we know that function f is positive and increasing, and the function g is negative and decreasing. Determine whether the following functions are increasing or decreasing.

- (a) The product function: $f(x)g(x)$.
- (b) The composite function: $f(g(x))$.

Problem 3

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Problem 3

Problem 3 Let $f(x) = 2x^3 - 4x^2 + 1151$ and $g(x) = x^3 \sin(x)$. Compute $f^{(4)}(x)$ and $g^{(4)}(x)$.

Problem 4

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Problem 4

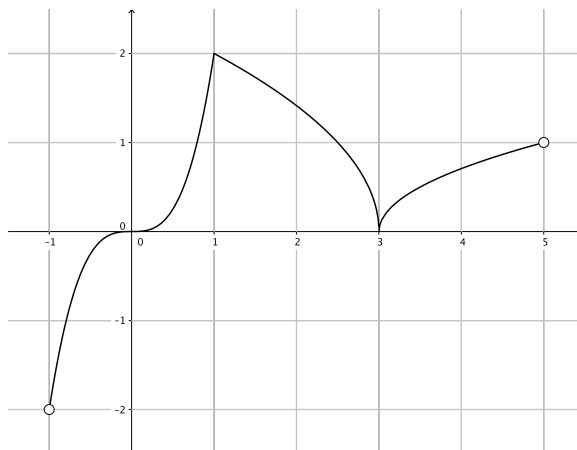
Problem 4 Let $f(x) = \sin(x)$ and $g(x) = \cos(x)$. Can you compute $f^{(48)}(x)$, $g^{(42)}(x)$, and $g^{(39)}(x)$?

Problem 5

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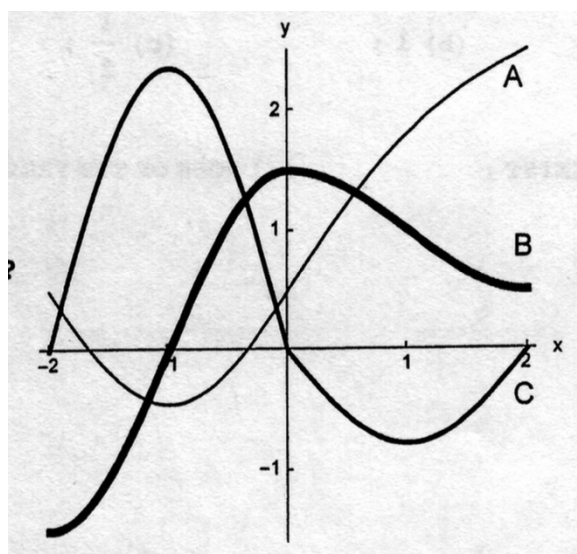
Problem 5

- Problem 5** (a) The (entire) graph of a function f is shown in the figure below.



Find the interval (or intervals) on which the derivative of f is increasing.

- (b) The figure below shows the graphs of f , f' , and f'' . Which curve is which?



Problem 6

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Problem 6

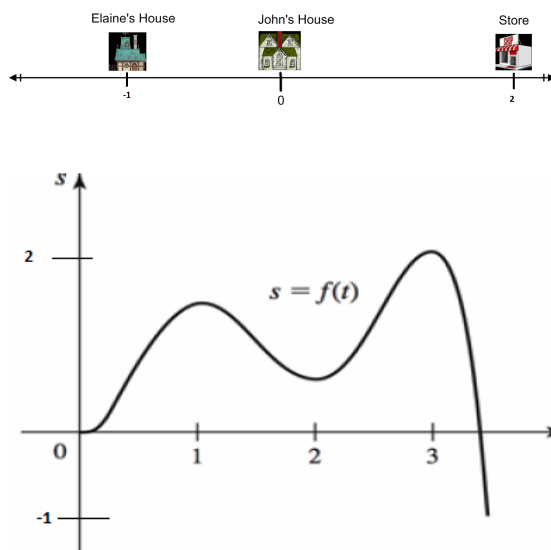
Problem 6 Consider the parabola $f(x) = ax^2 + bx + c$ where a, b, c are constants. For what values of a, b, c is f concave up? For what values of a, b, c is f concave down?

Problem 7

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Problem 7

Problem 7 John is walking along a straight path. His position at the time $t > 0$ is given by $s = f(t)$. He starts at $t = 0$ from his house ($f(0) = 0$) and the graph of f is given below.



- Describe the motion of John as precisely as you can.
- When is John's velocity zero? What is happening to John at those times?
- When is John moving in the positive direction? When is John the furthest in the positive direction and the furthest in the negative direction?
- When is John's velocity increasing? And when is it decreasing? When is John going at maximum velocity?

Problem 8

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Problem 8

Problem 8 Suppose that a stone is thrown vertically upward from a cliff on Mars with an initial velocity of 24 ft/s from a height of 192 ft. The height s of the stone above the ground after t seconds is given by $s(t) = -6t^2 + 24t + 192$.

- (a) Determine the velocity and acceleration of the stone after t seconds.
- (b) What is the greatest height of the stone and when does it occur? What are the velocity and acceleration at that time?
- (c) When does the stone hit the ground? What are the velocity and acceleration at that time?

Problem 9

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Problem 9

Problem 9 Suppose the function f has derivative given by $f'(x) = \frac{2x}{4+x^2}$.

- (a) Find a formula for $f''(x)$.
- (b) Construct a sign-chart for f' and f'' . [2]
- (c) On what interval(s) is f increasing? On what interval(s) is f concave down?

Problem 10

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Problem 10

- Problem 10** (a) *True or False: If the acceleration of an object is constant, then its velocity is constant.*
- (b) *True or False: A moving object can have negative acceleration and increasing speed.*

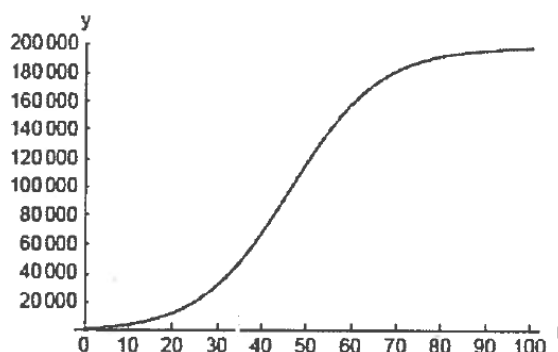
Problem 11

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Problem 11

Problem 11 The total number of people, N who have contracted a common cold by a time t days after its outbreak on an island is given by $N = N(t) = \frac{200000}{1 + 100e^{-0.1t}}, t \geq 0$.

- (a) Evaluate and interpret the limit $\lim_{t \rightarrow \infty} N(t)$
- (b) How long will it take for the number of people who have contracted the cold to reach 40,000?
- (c) The graph of the function N on the interval $[0, 100]$ is given below. Sketch (as best you can) the graph of its derivative, $N'(t)$.



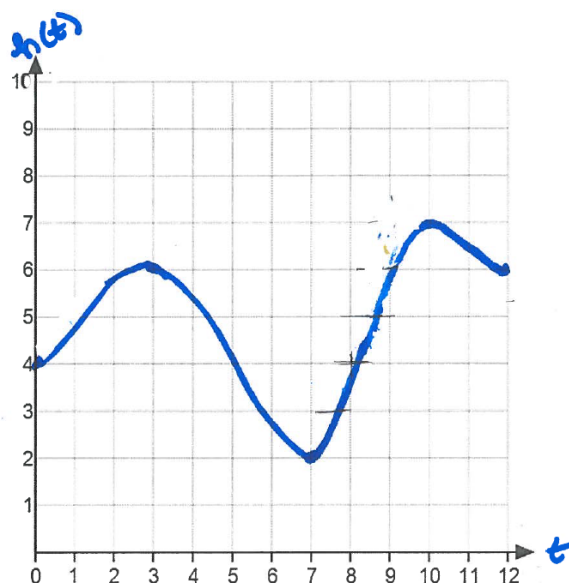
- (d) Calculate $N'(t)$. What does $N'(t)$ represent?
- (e) Evaluate and interpret the limit $\lim_{t \rightarrow \infty} N'(t)$
- (f) Find the average growth rate of the number of people who have contracted the disease during the time interval $[5, 6]$ (or during the sixth day after the outbreak).
- (g) Find the instantaneous growth rate of the number of people who have contracted the disease for $t = 5$ (round to a whole number).

Problem 12

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Problem 12

Problem 12 The graph of $h(t)$ represents the height in feet of water in a pool at time t minutes.



- During what time intervals is the height of water in the pool increasing? Decreasing?
- Assume the rate of change of the water in the pool is 0 when $t = 0$. For what other values of t is the rate of change of the water in the pool 0?
- If $h(t)$ represents the height of the feet in water at time t minutes. What function represents the rate of change of the height over time?
- Sketch a graph of the rate of change of the height over time, $h'(t)$.